CSIDNet: Compact single-image dehazing network for outdoor scene enhancement

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***Abstract*—Single-image dehazing is significantly important for improving the performance of outdoor scenes affected by atmospheric degradation. In this work, we have presented a compact deep-learning-based framework named as "The Compact Single Image Dehazing Network" (CSIDNet), designed to directly generate dehazed images from a single hazy input image. The proposed model consists of only three convolutional layers, thereby significantly reducing computational complexity as well as the training requirements.**

Unlike the conventional deep models that require large-scale datasets, CSIDNet achieves great performance using a relatively small training set without compromising the restoration quality. The model is trained on both synthetic and real-world images from the widely used RESIDE dataset, including the Outdoor Training Set (OTS), Synthetic Objective Testing Set (SOTS).

Evaluation using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) shows that the proposed method performs comparatively or better than several existing approaches. CSIDNet is well-suited for real-time applications such as autonomous driving, surveillance, where based on a single image the machine has to make a decision and if there is haze in the scene then single-image dehazing method proposed in this paper can improve it.

***Keywords*—Image Dehazing, Residual Learning, CNN, Dark Channel, Illumination Channel**

I. Introduction

Images captured in outdoor environments often suffer from degradation due to haze, fog, and other atmospheric conditions and aerosols. This degradation is primarily caused by scattering and absorption of light, leading to reduced visibility and contrast.

The degradation model is typically represented as:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where $I(x)$ is the observed hazy image, $J(x)$ is the scene radiance, $t(x)$ is transmission, and A is atmospheric light.

Although traditional methods attempt to estimate $t(x)$ and A, these approaches often fail in real-world scenarios.

Deep learning models have shown promising results; however, they are often computationally heavy. Therefore, there is a need for lightweight yet effective architectures.

II. Motivation

The key motivations behind this work are to :

- Reduce computational complexity of deep learning-based dehazing models
- Achieve real-time performance with minimal training data

III. Related Work

Traditional methods such as the Dark Channel Prior (DCP) rely on handcrafted priors but fail in sky regions. The drawback of DCP is haze overestimation in the sky. Learning-based approaches such as DehazeNet and AOD-Net improved performance but require large datasets. Recent works have explored attention mechanisms and feature fusion; however, these approaches increase model complexity. This motivates the design of a compact yet effective architecture.

IV. Proposed Methodology

A. Overall Pipeline

The proposed framework consists of three major stages:

1) Feature Extraction (includes 2 channels)
2) Feature Fusion and Residual Learning
3) Image Reconstruction

Fig. 1. Output of an image using CSIDNet

B. Feature Extraction

1) Dark Channel Prior: The dark channel helps in identifying dense haze regions:

$$I_{dark}(x) = \min_{c \in \{R,G,B\}} \left(\min_{y \in \Omega(x)} I^c(y) \right) \tag{2}$$

2) Illumination Channel: The illumination channel captures brightness variations:

$$I_Y = 0.299R + 0.587G + 0.114B \tag{3}$$

3) Feature Stack: The final input tensor is:

$$I_{input} = [R, G, B, I_{dark}, I_Y] \tag{4}$$

C. Feature Fusion Module

To combine extracted features correctly, we use a convolution-based fusion method:

$$F = \sigma(W_f * I_{input}) \tag{5}$$

Batch normalization and Leaky ReLU are applied to stabilize the training process.

D. Residual Learning

Instead of directly producing the clean image, the network learns residual haze present in the image:

$$R(x) = I(x) - J(x) \tag{6}$$

The restored image is obtained as:

$$J(x) = I(x) - R(x) \tag{7}$$

This mathematical formula improves convergence and preserves fine details.

E. Network Architecture

The architecture consists of:

- Initial feature extraction layer
- Three Convolutional layers
- Residual connection
- Sigmoid output layer

1) Layer-wise Representation:

$$F_{l+1} = \sigma(BN(W_l * F_l)) \tag{8}$$

2) Skip Connection:

$$F_{out} = F_{input} + F_{deep} \tag{9}$$

F. Loss Function

We use a hybrid loss:

$$L = \lambda L_{MSE} + (1 - \lambda) L_{L1} \tag{10}$$

$$L_{MSE} = \frac{1}{N} \sum (I_{pred} - I_{gt})^2 \tag{11}$$

This ensures both pixel accuracy and structural similarity.

V. Dataset

The model is trained and evaluated on the RESIDE dataset, which includes:

- OTS (Training)
- SOTS (Testing)
- HSTS (Real-world testing)

VI. Experimental Setup

A. Training Details

- Optimizer: Adam
- Learning rate: 0.0001
- Epochs: 100
- Batch size: 8

B. Evaluation Metrics

- PSNR (Peak Signal to Noise Ratio)
- SSIM (Structural Similarity Index)

VII. Results and Validation

A. Quantitative Results

TABLE I
Performance Comparison

Method	PSNR	SSIM	Time (s)
DCP	16.6	0.81	High
AOD-Net	28.2	0.90	Medium
Proposed	29.1	0.88	Low

B. Qualitative Results

Fig. 2. Visual comparison

C. Ablation Study

TABLE II
Effect of Features

Features Used	PSNR	SSIM
RGB only	27.5	0.82
RGB + Dark	28.5	0.86
RGB + Dark + Illumination	29.1	0.88

VIII. Discussion

The experimental results showed that the proposed model performs effectively in different configurations. The use of three convolutional layers is to provide the best balance between performance and computational efficiency. Increasing the number of layers or filters does not significantly improve performance instead increases training time.

The model was trained on small subsets of the dataset (200 OTS images), and consistent results were obtained in different training sets. This indicates that the network is stable and does not depend heavily on large amounts of data.

Overall the compact architecture combined with feature augmentation and residual learning, helps in achieving good visual quality while maintaining low computational cost. This makes the proposed model suitable for real-time applications.

IX. Conclusion

In this paper, we have proposed a lightweight feature-augmented residual network for single image dehazing. The model effectively improves image clarity with low computational cost. Experimental results show that it achieves a good balance between performance and efficiency, which makes it suitable for real-time applications.

References

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